UNNI CO2 SMARTIFICATION — REV A

Two-sheet cleanup: standard power symbols, grid-aligned wiring, explicit local connectivity.

USB + Power

File: 01_usb_power.kicad_sch

ESP32-C6 + Auxiliary

File: 02_esp_aux.kicad_sch

Simulation Harness

File: 03_simulation_harness.kicad_sch

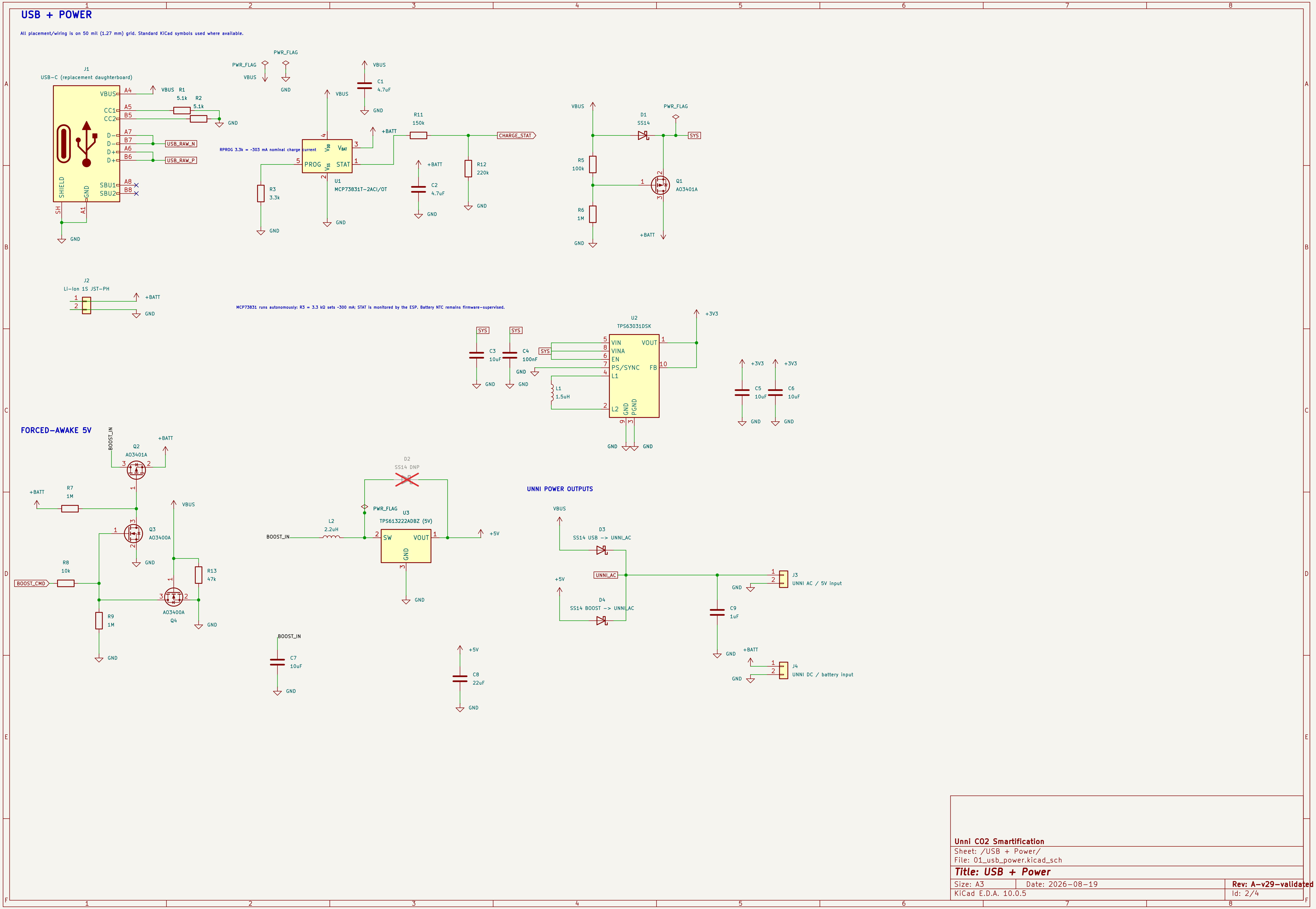
Unni CO2 Smartification

Sheet: /

File: unni-smartification-c6.kicad_sch

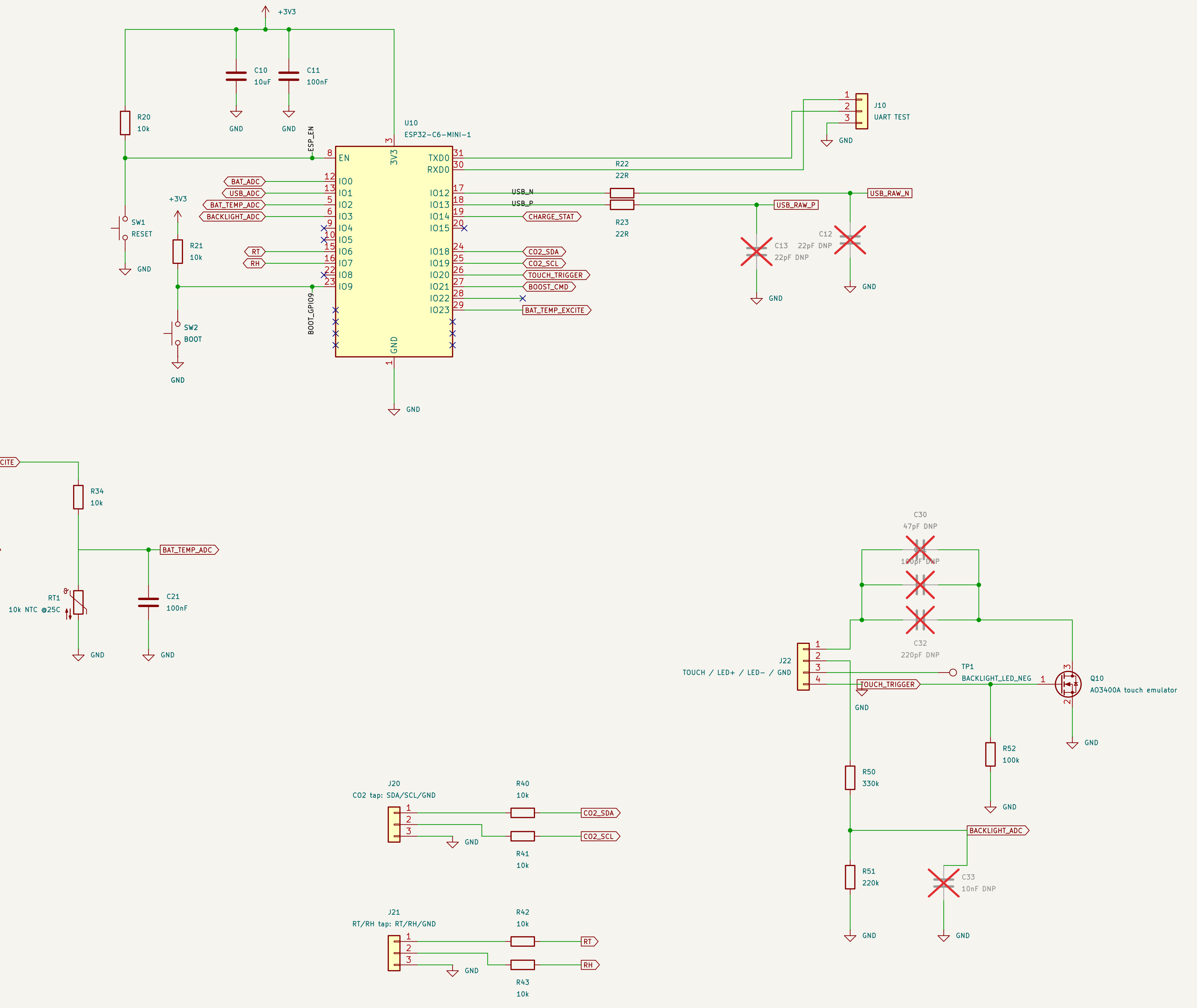
Title: Unni CO2 Smartification — ESP32-C6 Rev A

Size: A3	Date: 2026-08-19	**Rev: A-v29-validated**
KiCad E.D.A. 10.0.5		Id: 1/4


USB + POWER
All placement/wiring is on 50 mil (1.27 mm) grid. Standard KiCad symbols used where available.
J1
USB-C (replacement daughterboard)
VBUS
CC1
CC2
D−
D−
D+
D+
SBU1
SBU2
SHIELD
GND
A4
A5
B5
A7
B7
A6
B6
A8
B8
SH
A1
R1
5.1k
R2
5.1k
USB_RAW_N
USB_RAW_P
GND
PWR_FLAG
VBUS
RPROG 3.3k = ~303 mA nominal charge current
PROG
STAT
U1
MCP73831T-2ACI/OT
R3
3.3k
C1
4.7uF
+BATT
C2
4.7uF
R11
150k
R12
220k
CHARGE_STAT
D1
SS14
SYS
R5
100k
R6
1M
Q1
AO3401A
J2
Li-Ion 1S JST-PH
MCP73831 runs autonomously: R3 = 3.3 kΩ sets ~300 mA; STAT is monitored by the ESP. Battery NTC remains firmware-supervised.
U2
TPS63031DSK
VIN
VINA
EN
PS/SYNC
L1
L2
VOUT
FB
PGND
+3V3
C3
10uF
C4
100nF
L1
1.5uH
C5
10uF
C6
10uF
FORCED-AWAKE 5V
BOOST_IN
Q2
AO3401A
R7
1M
Q3
AO3400A
R8
10k
BOOST_CMD
R9
1M
R13
47k
AO3400A
Q4
D2
SS14 DNP
L2
2.2uH
U3
TPS613222ADBZ (5V)
SW
VOUT
+5V
C7
10uF
C8
22uF
UNNI POWER OUTPUTS
D3
SS14 USB -> UNNI_AC
D4
SS14 BOOST -> UNNI_AC
UNNI_AC
C9
1uF
J3
UNNI AC / 5V input
J4
UNNI DC / battery input
Unni CO2 Smartification
Sheet: /USB + Power/
File: 01_usb_power.kicad_sch
Title: USB + Power
Size: A3
Date: 2026-08-19
KiCad E.D.A. 10.0.5
Rev: A-v29-validated
Id: 2/4

ESP32-C6 + AUXILIARY CIRCUITS

GPIO0-3 are reserved for ADC inputs; RT/RH use RTC-capable GPIO6/7 for deep-sleep wake.



Unni CO2 Smartification

Sheet: /ESP32-C6 + Auxiliary/

File: 02_esp_aux.kicad_sch

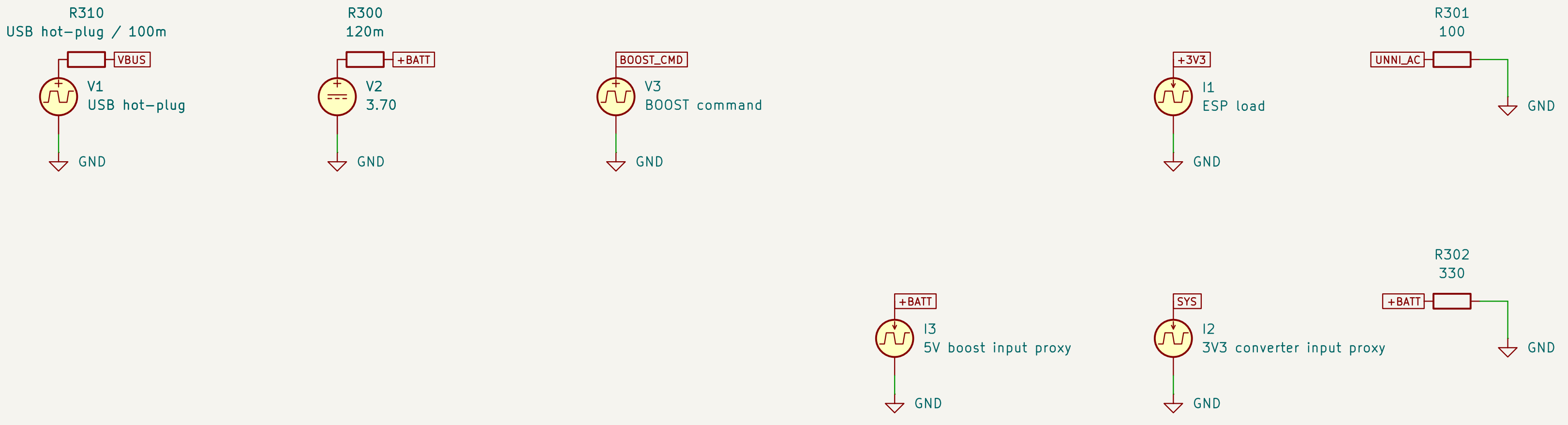
Title: ESP32-C6 + Auxiliary Circuits

Size: A3 | Date: 2026-08-19 | Rev: A-v29-validated

KiCad E.D.A. 10.0.5 | Id: 3/4

SIMULATION HARNESS — excluded from BOM/PCB

Scenario: battery−only → forced awake → USB hot−plug while BOOST_CMD remains asserted → autonomous charging → USB removal. Battery model includes 120 mΩ ESR. Battery model includes 120 mΩ ESR. I2 is the 3V3−converter input−power proxy. TPS613222A input power/discharge is modeled internally; legacy I3 is excluded from simulation.



Unni CO2 Smartification

Sheet: /Simulation Harness/

File: 03_simulation_harness.kicad_sch

Title: Simulation Harness

Size: A3 | Date: 2026-08-19 | Rev: A-v29-validated

KiCad E.D.A. 10.0.5 | Id: 4/4